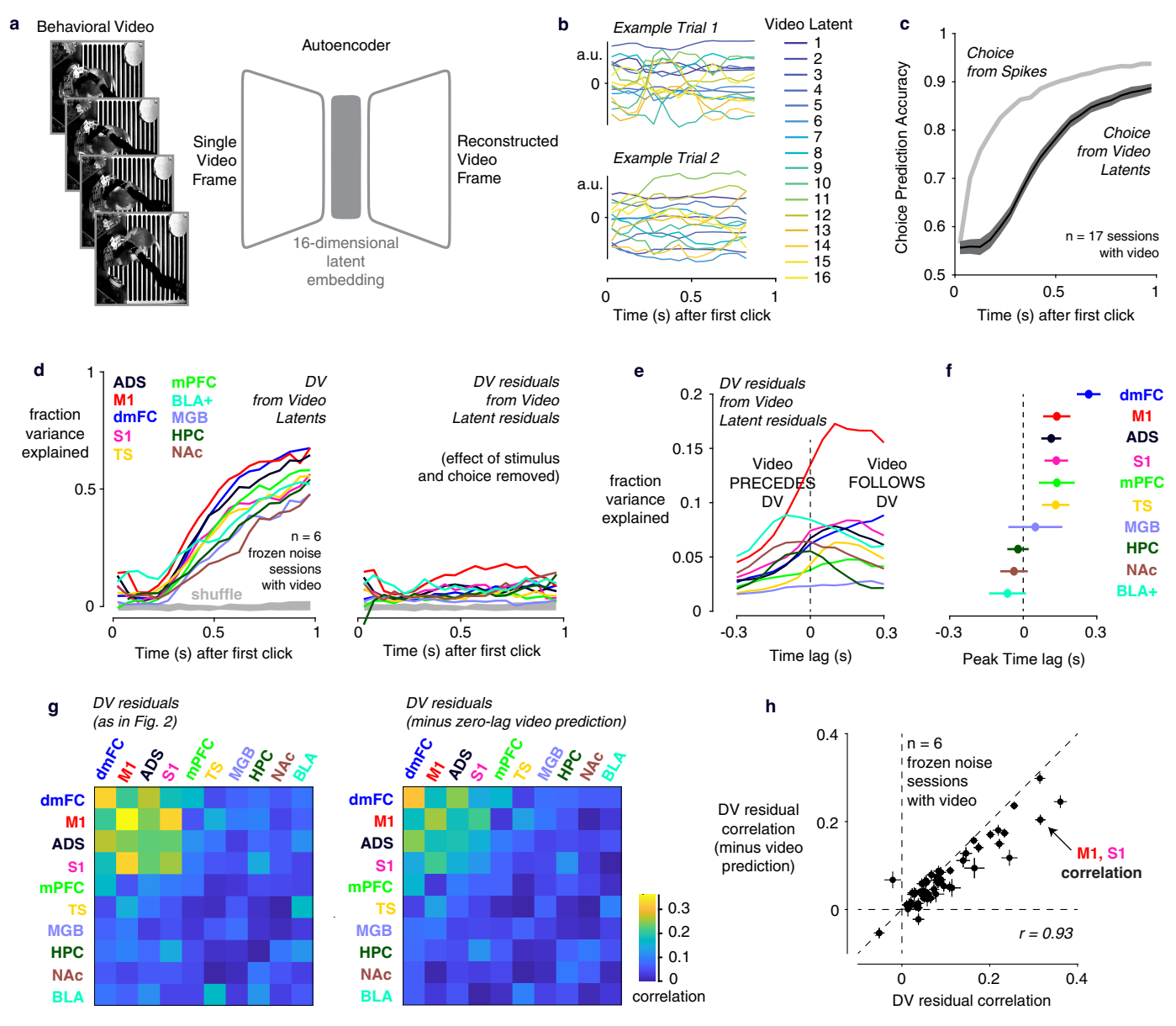

a
Behavioral Video
Autoencoder
Single Video Frame
Reconstructed Video Frame
16-dimensional latent embedding
b
Example Trial 1
Example Trial 2
a.u.
Video Latent
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
Time (s) after first click
c
Choice from Spikes
Choice from Video Latents
Choice Prediction Accuracy
n = 17 sessions with video
Time (s) after first click
d
fraction variance explained
ADS
M1
dmFC
S1
TS
mPFC
BLA+
MGB
HPC
NAc
DV from Video Latents
DV residuals from Video Latent residuals
(effect of stimulus and choice removed)
n = 6 frozen noise sessions with video
shuffle
Time (s) after first click
e
DV residuals from Video Latent residuals
Video PRECEDES DV
Video FOLLOWS DV
fraction variance explained
Time lag (s)
f
dmFC
M1
ADS
S1
mPFC
TS
MGB
HPC
NAc
BLA+
Peak Time lag (s)
g
DV residuals (as in Fig. 2)
DV residuals (minus zero-lag video prediction)
correlation
h
n = 6 frozen noise sessions with video
M1, S1 correlation
r = 0.93
DV residual correlation (minus video prediction)
DV residual correlation